

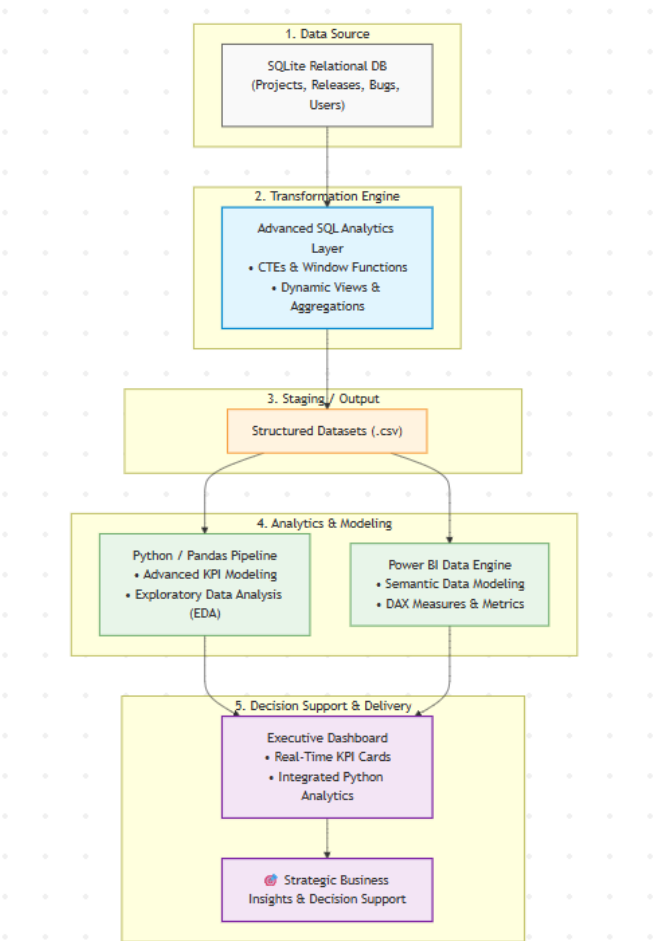
Project Architecture

Release Quality Analytics Platform

Purpose

This document presents the end-to-end architecture of the **Release Quality Analytics Platform**. It demonstrates how raw software testing data is collected, transformed, analyzed, and visualized using SQLite, SQL, Python, DAX, and Power BI.

End-to-End Architecture



Technologies

Layer	Technology
Database	SQLite
Query Language	SQL
Data Processing	Python (Pandas)
Business Intelligence	Power BI
Data Modeling	DAX
Documentation	Markdown & PDF

Data Flow

1. Raw software testing data is stored in the SQLite database.
 2. SQL queries extract and transform the data using JOIN, GROUP BY, CASE, CTEs, Window Functions, Views, and Aggregate Functions.
 3. Query outputs are exported as CSV files.
 4. Python performs additional analytical calculations and KPI generation.
 5. Power BI imports both SQL outputs and Python-generated datasets.
 6. DAX measures calculate dynamic KPIs and business metrics.
 7. Interactive dashboards present actionable insights for Product Owners, QA Engineers, Business Analysts, and Project Managers.
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Key Features

- End-to-end analytics workflow
- Relational database design
- SQL-based data transformation
- Python analytics
- DAX calculations
- Interactive Power BI dashboards
- Business-oriented reporting